

A 36-point subset of the $13 \times 13 \times 13$ grid with no five points on a sphere or a plane

John Erlbacher*

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Abstract

For the “no 5 on a sphere” problem — determine $C(n)$, the size of the largest subset of the grid $\{1, \dots, n\}^3$ containing no five points on a common sphere or plane — the strongest published lower bounds for $7 \leq n \leq 12$ are due to AlphaEvolve, ending with $C(12) \geq 33$; nothing specific to $n = 13$ has been published, so the best previously known lower bound was $C(13) \geq C(12) \geq 33$. We exhibit an explicit 36-point subset of $\{0, \dots, 12\}^3$ with no five points on a sphere or a plane, proving $C(13) \geq 36$. The set is invariant under the central inversion $p \mapsto (12, 12, 12) - p$: eighteen antipodal point pairs whose squared distances from the cube center are pairwise distinct. Verification is a finite exact-integer computation — all $\binom{36}{5} = 376,992$ lifted 5×5 determinants are nonzero — reproduced here by a 21-line program, printed in full, and carried out by three independently written checkers validated by mutation testing. The set was found in 841 seconds on eight CPU threads by an exact-arithmetic local search restricted to centrally symmetric configurations; en route, four independent search pilots produced thousands of distinct 34-point sets and dozens of 35-point sets. The trivial upper bound $C(13) \leq 52$ is unaffected.

1 Introduction

The following problem appears as Problem 6.60 in the AlphaEvolve “repository of problems” of Georgiev, Gómez-Serrano, Tao and Wagner [5, 6]:

“For n a natural number, let $C(n)$ denote the size of the largest subset of $[n]^3 = \{1, \dots, n\}^3$ such that no 5 points lie on a sphere or a plane. Obtain upper and lower bounds for $C(n)$ that are as strong as possible.”

It is the three-dimensional case of the no- $(d+2)$ -on-a-sphere problem (see [2, Problem 4 of Chapter 10.1]), a generalization, going back to Thiele [10], of the no-four-on-a-circle problem of Erdős and Purdy. Thiele [10] proved $c\sqrt{n} \leq C(n) \leq 4n$ (the upper bound is the observation that each of the n planes $x = \text{const}$ carries at most 4 points of a valid set); Suk and White [8] improved the lower bound to $n^{3/4-o(1)}$, and Dong and Xu [4] to $n - o(n)$, with a related construction from rational normal curves given by Szabó [9]. These constructions are asymptotic and yield no values for specific small n .

For small n the record lower bounds come from AI-driven search. PatternBoost [3] reported $C(3) \geq 8$, $C(4) \geq 11$, $C(5) \geq 14$, $C(6) \geq 18$, $C(7) \geq 20$, $C(8) \geq 22$, $C(9) \geq 25$ and $C(10) \geq 27$;¹ AlphaEvolve [7, 5] improved these to $C(7) \geq 21$, $C(8) \geq 23$, $C(9) \geq 26$, $C(10) \geq 28$, and added the new bounds $C(11) \geq 31$ and $C(12) \geq 33$ [5, Problem 6.60]. Beyond the general bounds above and the monotone consequence $C(13) \geq C(12) \geq 33$, we could find no published bound for any specific $n \geq 13$ (literature and web searches dated 11–12 June 2026; see Section 5), so the baseline for $n = 13$ was 33. This note establishes:

Theorem 1.1. *$C(13) \geq 36$. Explicitly, the 36-point set $S \subseteq \{0, \dots, 12\}^3$ of Table 1 contains no five points on a common sphere or plane.*

*Independent Researcher. Email: erlbacher.research@gmail.com. ORCID: 0009-0003-6851-4139.

¹The explicit witnesses printed in the appendix of [3] pass the determinant criterion of Proposition 2.1 for $n = 3, \dots, 9$ (we re-checked them), but the printed $n = 10$ list is corrupted — as printed it contains coordinates up to 13 and a vanishing 5-subset — so we quote that value as reported; it is in any case superseded by AlphaEvolve’s verified $C(10) \geq 28$.

By monotonicity, $C(n) \geq 36$ for all $n \geq 13$. The best known upper bound remains the trivial $C(13) \leq 4 \cdot 13 = 52$.

Conventions. We work in $\{0, \dots, 12\}^3$; adding 1 to every coordinate places S in the problem’s $[13]^3$, and the defining condition is translation-invariant. “No five on a sphere or a plane” is interpreted, as in the source problem and its official evaluation code [6], through the determinant criterion of Proposition 2.1, which treats planes as degenerate spheres; the checkers used here accept all six published AlphaEvolve record sets for $n = 7, \dots, 12$ under this criterion (Section 3), so the comparison $33 \rightarrow 36$ is made under the very convention that produced the 33.

Methods and AI-use disclosure. The set was found by Demonstrandum, a verification-first multi-agent AI search pipeline built on Claude language models (Anthropic): agents froze the problem statement and conventions from the official sources, proposed attack plans, and wrote and ran the search programs (Rust and Python) of Section 4; every geometric test in the search and every step of the verification is exact integer arithmetic (floating point appears only in search-heuristic branch probabilities and timers, never in any validity decision). Certificates were verified by three independently written exact-integer checkers, validated by mutation testing, plus a fourth written by an adversarial referee agent from a different model family (Codex, GPT-5.5, OpenAI) (Section 3). The text of this note was likewise drafted with the Claude-based pipeline; the author directed the work and takes responsibility for its content. Every number in this note that derives from the certificate or from our own runs was recomputed from the underlying artifacts during its preparation (12 June 2026). The mathematical claim of Theorem 1.1 rests solely on the explicit set of Table 1 and the finite computation of Section 3, which any reader can reproduce in under a minute.

2 The certificate

Throughout, identify a point $p = (x, y, z) \in \mathbb{R}^3$ with its *lift* $\Lambda(p) = (x, y, z, x^2+y^2+z^2, 1) \in \mathbb{R}^5$.

Proposition 2.1. *Five pairwise distinct points $p_1, \dots, p_5 \in \mathbb{R}^3$ lie on a common sphere or plane if and only if $\det(\Lambda(p_1), \dots, \Lambda(p_5)) = 0$.*

Proof. A sphere or plane is the zero set of $f(x, y, z) = a(x^2+y^2+z^2) + bx + cy + dz + e$ for some $(a, b, c, d, e) \neq 0$: a plane when $a = 0$ (then $(b, c, d) \neq 0$, else $f \equiv e \neq 0$ has empty zero set), a sphere of radius r , $r^2 = (b^2+c^2+d^2-4ae)/4a^2$, when $a \neq 0$. If the five points lie on such a zero set, the vector (b, c, d, a, e) is orthogonal to all five lifts, so the determinant vanishes. Conversely, if the determinant vanishes, some $(b, c, d, a, e) \neq 0$ is orthogonal to all five lifts, giving an f as above vanishing at all five points; if $a \neq 0$, its zero set contains five distinct points, hence is neither empty nor a single point, hence is a genuine sphere ($r^2 > 0$); if $a = 0$ it is a plane. $\square$

Call a finite set $S \subset \mathbb{R}^3$ *valid* if every 5-subset has nonvanishing lifted determinant. By Proposition 2.1 this is exactly the problem’s condition. One consequence used heavily by the search (but not needed for verification) is worth recording:

Lemma 2.2. *If four distinct points are collinear, or lie on a common circle, then these four points together with any fifth point have vanishing lifted determinant.*

Proof. A line is contained in two distinct planes, and a circle in a plane and a sphere; in both cases the four lifts satisfy two linearly independent orthogonality constraints, so they span a subspace of dimension at most 3. Together with the lift of any fifth point they therefore lie in a subspace of dimension at most 4, and the 5×5 determinant vanishes. $\square$

Thus a valid set with at least 5 points has at most 3 points on any line and at most 3 on any circle, and a search must treat such *rank-degenerate* quadruples as blocking every cell.

The certificate set S is reproduced in machine-readable form here:

```
[0,1,12],[0,3,7],[0,9,10],[0,12,0],[1,4,1],[1,6,4],[1,12,11],[2,6,12],
[2,8,1],[3,1,10],[3,5,7],[3,7,10],[4,3,5],[4,8,3],[5,0,0],[5,2,2],
[5,11,3],[6,0,4],[6,12,8],[7,1,9],[7,10,10],[7,12,12],[8,4,9],[8,9,7],
[9,5,2],[9,7,5],[9,11,2],[10,4,11],[10,6,0],[11,0,1],[11,6,8],[11,8,11],
[12,0,12],[12,3,2],[12,9,5],[12,11,0]]
```

#	p	$\sigma(p)$	$4\rho^2$	#	p	$\sigma(p)$	$4\rho^2$
1	(3, 5, 7)	(9, 7, 5)	44	10	(0, 3, 7)	(12, 9, 5)	184
2	(4, 3, 5)	(8, 9, 7)	56	11	(3, 1, 10)	(9, 11, 2)	200
3	(4, 8, 3)	(8, 4, 9)	68	12	(2, 6, 12)	(10, 6, 0)	208
4	(3, 7, 10)	(9, 5, 2)	104	13	(1, 4, 1)	(11, 8, 11)	216
5	(1, 6, 4)	(11, 6, 8)	116	14	(0, 9, 10)	(12, 3, 2)	244
6	(5, 2, 2)	(7, 10, 10)	132	15	(5, 0, 0)	(7, 12, 12)	292
7	(5, 11, 3)	(7, 1, 9)	140	16	(1, 12, 11)	(11, 0, 1)	344
8	(6, 0, 4)	(6, 12, 8)	160	17	(0, 1, 12)	(12, 11, 0)	388
9	(2, 8, 1)	(10, 4, 11)	180	18	(0, 12, 0)	(12, 0, 12)	432

Table 1: The certificate: a valid 36-point set $S \subset \{0, \dots, 12\}^3$, listed as the 18 orbit pairs $\{p, \sigma(p)\}$ of the central inversion $\sigma(p) = (12, 12, 12) - p$, sorted by the squared diameter $4\rho^2 = 4|p - c|^2 \in \mathbb{Z}$ of the pair about the cube center $c = (6, 6, 6)$. The 18 values $4\rho^2$ are pairwise distinct (Lemma 2.3 (ii) explains why they must be).

Structure. S is invariant under the central inversion $\sigma(p) = (12, 12, 12) - p$ of the 13-cube, with 18 orbit pairs and no fixed point (the unique fixed cell $(6, 6, 6)$ is not in S — by Lemma 2.3 (i) it could not be). The per-axis layer profiles (number of points with the given coordinate value $0, 1, \dots, 12$), palindromic by symmetry, are

$$x: 4, 3, 2, 3, 2, 3, 2, 3, 2, 3, 2, 3, 4; \quad y: 4, 3, 1, 3, 3, 2, 4, 2, 3, 3, 1, 3, 4; \quad z: 4, 3, 4, 2, 2, 3, 0, 3, 2, 2, 4, 3, 4;$$

so every axis-aligned plane carries at most 4 points — the budget that the trivial bound $C(13) \leq 52$ counts. 16 of the 36 points lie on the surface of the cube. Over all 376,992 lifted 5×5 determinants the minimum absolute value is 2 (200 subsets have $|\det| \leq 10$; the maximum is 198,750): like the published record sets for $n \leq 12$, the object sits at the very edge of degeneracy. Finally, S is *saturated*: an exhaustive check of all $2197 - 36$ remaining cells of $\{0, \dots, 12\}^3$ (re-run while preparing this note) shows that no 37th point can be added.

The following elementary facts drove the choice of search space; they are stated for the 13-cube but hold for any odd n .

Lemma 2.3. *Let $c = (6, 6, 6)$ and $\sigma(p) = 2c - p$, and let T be a σ -invariant subset of $\{0, \dots, 12\}^3$.*

(i) *If $c \in T$ and T contains at least two orbit pairs, then T is not valid.*

(ii) *If two orbit pairs $\{p, \sigma(p)\}$, $\{q, \sigma(q)\}$ of T satisfy $|p - c| = |q - c|$, then these four points are cocircular or collinear, so (by Lemma 2.2) T is not valid if $|T| \geq 5$.*

(iii) *Any two orbit pairs of T form a parallelogram, hence a coplanar quadruple.*

Proof. (iii): the diagonals $[p, \sigma(p)]$ and $[q, \sigma(q)]$ bisect each other at c . (ii): a parallelogram with equal diagonals is a rectangle (or a degenerate one: four collinear points), and a rectangle is cyclic. (i): let $\{p, \sigma(p)\}$ and $\{q, \sigma(q)\}$ be two orbit pairs of T . If p, q and c are collinear, all five points lie on one line, hence on one plane. Otherwise the plane through c, p and q is σ -invariant (it contains c), so it also contains $\sigma(p)$ and $\sigma(q)$: again five points of T on one plane. $\square$

3 Verification

Theorem 1.1 reduces, via Proposition 2.1, to a finite integer computation: all $\binom{36}{5} = 376,992$ five-subsets of S have nonzero lifted determinant. Subtracting the first row of the 5×5 matrix from the others and expanding along the last column reduces this to a 4×4 integer determinant in the differences $\Lambda'(p_i) - \Lambda'(p_1)$, $\Lambda'(p) = (x, y, z, x^2 + y^2 + z^2)$. Coordinate differences are at most 12 and lifted fourth-coordinate differences at most 432 in absolute value, so every such determinant is bounded by $24 \cdot 12^3 \cdot 432 = 17,915,904 < 2^{25}$: native 64-bit (even 32-bit) integer arithmetic is exact, and no floating point is needed or used. The complete accept path of the reference checker is the following 21 lines of Python (quoted verbatim from `check_cert.py` in the certificate bundle):

```

import sys, json
from itertools import combinations

def check(points, n=13):
    """Return (True, None) if valid, else (False, reason-or-offending-5-subset)."""
    pts = [tuple(p) for p in points]
    if len(set(pts)) != len(pts):
        return False, "duplicate points"
    if any(len(p) != 3 or any(not isinstance(c, int) or not 0 <= c < n for c in p) for p in pts):
        return False, "non-integer or out-of-range coordinate"
    L = [(x, y, z, x * x + y * y + z * z) for (x, y, z) in pts]
    for idx in combinations(range(len(pts)), 5):
        p = L[idx[0]]
        (a0, a1, a2, a3), (b0, b1, b2, b3), (c0, c1, c2, c3), (d0, d1, d2, d3) = \
            [tuple(L[i][k] - p[k] for k in range(4)) for i in idx[1:]]
        det = ((a0*b1 - a1*b0) * (c2*d3 - c3*d2) - (a0*b2 - a2*b0) * (c1*d3 - c3*d1)
              + (a0*b3 - a3*b0) * (c1*d2 - c2*d1) + (a1*b2 - a2*b1) * (c0*d3 - c3*d0)
              - (a1*b3 - a3*b1) * (c0*d2 - c2*d0) + (a2*b3 - a3*b2) * (c0*d1 - c1*d0))
        if det == 0:
            return False, [pts[i] for i in idx]
    return True, None

```

Running `check_cert.py` on the list above prints `VALID m=36 n=13` in about a second. A skeptical reader is encouraged to re-implement the criterion independently rather than trust any of our code; the float-based verifier in the official problem notebook [6] should *not* be used as a certificate checker.

Beyond the reference checker (route A), the certificate bundle contains two independently written checkers: route B performs fraction-free Bareiss elimination on the full 5×5 lifted matrix (Python), and route C is a separate Rust implementation (with 28 unit tests, including traps built from the rank-degenerate quadruples of Lemma 2.2). All three routes were written by different agents than the one that found the set, and a fourth checker was written from scratch by an adversarial referee agent from a different model family (Codex, GPT-5.5), which confirmed all 376,992 determinants nonzero, minimum $|\det| = 2$, and the central symmetry. The checkers were validated by mutation testing: seven targeted corruptions of the certificate (a duplicated point; an out-of-range coordinate 13; a negative coordinate; a float coordinate; appending the center cell (6,6,6); replacing a point by the center; appending an in-range cell completing a coplanar 5-subset) are rejected by every applicable route: the float corruption is rejected by both Python routes and does not even parse for the integer-only Rust route; the other six corruptions are rejected by all three routes. As a convention anchor, the same checkers accept all six official AlphaEvolve record sets for $n = 7, \dots, 12$ (sizes 21, 23, 26, 28, 31, 33, taken verbatim from the official notebook [6]) and reject a known-invalid 34-point set. All of these checks, and every certificate- and run-derived number quoted in this note, were re-run on 12 June 2026 while preparing it.

4 Method: search in the centrally symmetric subspace

The published 33-point record set for $n = 12$ is the residue of a generic evolved local search [5, Problem 6.60], with no visible structure. Our route to 36 was to *impose* structure: the cube $\{0, \dots, 12\}^3$ has a true center $c = (6, 6, 6)$, and we searched only sets invariant under the central inversion $\sigma(p) = 2c - p$, i.e. unions of orbit pairs $\{p, \sigma(p)\}$. This halves the search dimension (36 points are 18 pair choices); by Lemma 2.3 (iii) every two orbit pairs automatically form a *coplanar, legal* quadruple, pre-paying densely the four-points-per-plane budget that record-grade sets must exhaust, while Lemma 2.3 (ii) gives a cheap exact pruning rule (one squared norm per pair) against the rectangle degeneracies symmetry would otherwise mass-produce.

The searcher (Rust; exact integer arithmetic in every geometric predicate) maintains an incrementally updated *blocker grid*: for the current set T and every cell u of the grid, the number of 4-subsets $Q \subseteq T$ whose lifted cofactor vector is orthogonal to $\Lambda(u)$; by Proposition 2.1, u is addable only if this count is zero. For a candidate *pair* $(p, \sigma(p))$, zero blocker counts at both cells decide all 5-subsets with at most one new point, and two further exact layers (cofactor tests over member 2-subsets, determinant tests over member 3-subsets) decide those containing both new points, including all rank-degenerate cases of Lemma 2.2. On top of this sits a standard iterated local search: greedy randomized fill over pair orbits, ruin-and-rebuild (remove 1–3

random pairs, rebuild with a tabu list, accept if not worse), restart after 800 non-improving iterations. Every accepted set is re-verified in-process by the brute-force all-subsets test before being written to disk.

bound	date found	pilot	evidence
$C(12) \geq 33$	Nov. 2025	AlphaEvolve (prior record)	[5, 6]
$C(13) \geq 34$	11 June 2026	centrally symmetric pilot, 4.0 s into its first smoke run (2 threads); independently baseline ILS, 53.8 s into its own smoke run; both further pilots followed within hours	run logs
$C(13) \geq 35$	11 June 2026	baseline ILS main run ($t = 538$ s); independently, blocker-repair pilot	run logs
$C(13) \geq 36$	12 June 2026	centrally symmetric main run: 8 threads, 840.9 s elapsed, 611,969 ILS iterations, 627 restarts	Table 1

Table 2: Record progression. All searches ran on one desktop CPU (Ryzen 9 9950X3D, 16 cores); roughly three hours elapsed between freezing the problem dossier and finding the 36-point set.

Four independent pilots ran in parallel (Tables 2 and 3): the centrally symmetric search; a faithful port of the published evolved $n = 12$ recipe (the official notebook discloses that, due to a misplaced `# EVOLVE-BLOCK-START` line, AlphaEvolve’s search for this problem “had to start its search from scratch every time” [6] — our port runs it both as published and with that harness bug fixed); an exact blocker-cofactor *repair* search over hitting sets of blocking quadruples; and a finite-field pilot mining mod-13 arc structure for seeds (a direct mod-13 construction of size 34 is impossible: arcs in $\text{PG}(4, 13)$ have at most 14 points by Ball’s theorem [1], so the lifted-matroid-mod-13 condition can serve only as seed diversity).

pilot	run length	distinct 34s	distinct 35s	36
baseline ILS (8 threads, two arms)	22 min snapshot	198	1	—
blocker-repair, 3 runs	55 + 48 + 48 min	4068 / 2528 / 2833	20 / 30 / 47	—
centrally symmetric (symmetric sets)	14 min	4126	n/a	1
finite-field $\mathbb{F}_{13}$, 2 seeded runs	5.5 + 11.8 min	69 / 71	—	—

Table 3: Distinct valid sets logged in the pilots’ principal runs (each set brute-force verified in-process before logging; counts deduplicated within each run by hash, not across runs and not up to cube symmetries). Of the baseline ILS 34s, 182 of 198 came from the arm with the harness bug fixed. A centrally symmetric set has even size, so the symmetric pilot cannot log a 35; its 34s are symmetric (17 pairs). The two finite-field runs’ 34-families overlapped almost completely: all 69 of the first run’s sets recur among the second’s 71.

Two findings frame the result. First, *abundance* (Table 3): valid 34-point sets in the 13-cube are not rare — pilots logged thousands of distinct ones within minutes, including 4126 distinct *centrally symmetric* 34-sets in the 14 minutes of the main symmetric run. Second, *saturation of the old record*: the published 33-point set admits no 34th point in the 13-cube — an exhaustive check of all eight translates of the 12-cube inside $\{0, \dots, 12\}^3$, re-run while preparing this note, finds zero addable cells for every translate. The published record was a dead local optimum; the 34–36 region was simply unexplored. Within the symmetric subspace the climb did not stop at 34: where the asymmetric pilots plateaued hard at 35 (the first blocker-repair run logged 510 dead-end local optima at size 34 and 18 at size 35 in 55 minutes), one symmetric run pushed through to 18 pairs, with the fixed seed 20260612, reproducible from the archived search-code snapshot and command line in the bundle.

5 What this does and does not show

- *The exact value of $C(13)$ remains unknown.* Theorem 1.1 is a lower bound; the best upper bound remains the trivial $C(13) \leq 52$. We make no optimality claim.
- *No empirical maximality claim either.* Our searches are heuristic. The specific set S admits no 37th grid point, but other 36-point sets, or larger sets, may exist in unexplored basins; no exhaustive search of any kind is claimed.

- *Nothing is claimed for other n .* In particular we did not find $C(12) \geq 34$, and we claim no bounds for $n = 14, 15$ beyond the monotone $C(n) \geq 36$.
- *No structural theorem.* Central symmetry is a search heuristic that worked, not a proven property of optimal sets.
- *Priority.* Openness was checked on 11–12 June 2026 against the official repository and paper [6, 5] (whose published data end at $C(12) \geq 33$), arXiv listings and API, GitHub code search, OpenReview, and general web search, including an independent sweep by the adversarial referee agent; no public claim of $C(13) \geq 34$ (or $C(12) \geq 34$) was found. Concurrent unpublished work cannot be excluded.

Data availability. The certificate (JSON), the three checkers and the referee’s independent checker, the mutation-test suite, the search code (Rust) with run logs and the discovering run’s command line, and full verification logs will be made available at [https://github.com/demonstrandum-research/\[REP0\]](https://github.com/demonstrandum-research/[REP0]). The certificate file has SHA-256 checksum

333d36ece36e3d845cd2f5bb26e5460f78d881c00418c92cae8bd5215ab0629a.

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